



FALCON's Response to the Requirement

TVSSCS: Pop-Up Divert Sorter

September 10, 2025

Doc. No. - Sales/FR/PT/007/R0



Kind Attention –

Mr. Santhanavadivel E

M/s TVS Supply Chain Solutions

Offer Ref: F24-00488-05; Date 10.09.2025

Subject –Techno-Commercial Offer for Pop-Up Divert Sorter

We are pleased to submit our detailed Techno-Commercial Offer in response to your RFP & our further discussions.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed techno-commercial proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to being your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

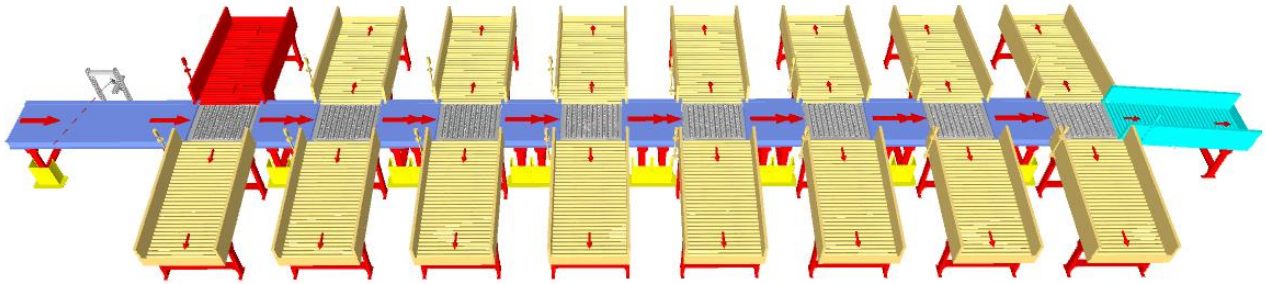
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the Proposal process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

Response to the Requirement for Pop-Up Divert Sorter



Proposal Reference: F24-00488-03

Falcon Autotech Private Limited

**Plot No. 87, Sector Ecotech-1, Extention-1, Greater Noida, Uttar
Pradesh 201308.**

Contact : Adib Hasim

Designation : Sr. Manager

Mob : +91 9711619998

Email : Adib.hasim@falconautotech.com

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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	AC	Alternating Current
4	DC	Direct Current
5	PLC	Programmable Logic Controller
6	IT	Information Technology
7	BOQ	Bill Of Quantity
8	PDP	Power Distribution Panel
9	PC	Personal Computer
10	UPS	Uninterrupted Power Supply

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to **TVSSCS** requirement for Swivel Wheel Divert Sorter. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.

Following the same objective for the Farukhnagar, Haryana, India system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **The system consists different Conveyors, Side Barcode Scanner & Pop-up Divert Units.**

A tailor-made and simple layout designed explicitly for TVSSCS. The proposed layout is the result of the technical requirements shared on emails and during our meetings with the relevant stakeholders.

1. Falcon's reliable Shipment sortation systems

Falcon Autotech has over 50+ Cross belt sorter loops installed globally which are used by most innovative brands such as Flipkart, Amazon, Delhivery, Aramex, Asendia, Fastway and many more. The main and critical components of the Falcon Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from industry leading suppliers in the world, such as SEW, Siemens, SICK, Vahle, Forbo etc. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the system's robustness and reliability.

2. Commitment to quality systems

- Demonstrating Falcon's clear commitment to the TVSSCS satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live and 1 Year Comprehensive Maintenance plan is provided as an option post warranty.
- A proven history of success and broad experience in implementing and maintaining automated systems.
- ISO 9001 Quality Assurance certification, ensuring consistent quality across our products, processes, and outcomes.
- Falcon Warehouse Control System (WCS) designed to meet the real-time operational needs of automated materials handling systems. WCS can be moulded basis the client's requirements to enhance operations.
- Comprehensive service and warranty programs. We deliver both operational and emergency support with unmatched responsiveness and availability throughout the region.
- Strong financial stability, backed by a long-standing track record.

3. A strong and already successful partnership between TVSSCS & Falcon

TVSSCS has been Falcon's customer since long and using multiple Falcon Systems. We have Arm Sorters installed at different locations. For each project, several advanced tools to improve the service's quality and efficiency are provided: Service Desk, Hotline support, remote access, governance meetings and monthly reports.

3. About Falcon

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0



Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces



Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS



Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5
Product Lines

15+
Countries with Live Installations

1800+
Total Installed Systems Globally

1Million+
Sorts Per Hour

600+
Employees

Long-standing Relationships with industry leaders

Ecom Retail

Flipkart

amazon

Lazada

مركز

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

SWIGGY

TITAN

Winkon India Pvt. Ltd.

News & Articles



FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.



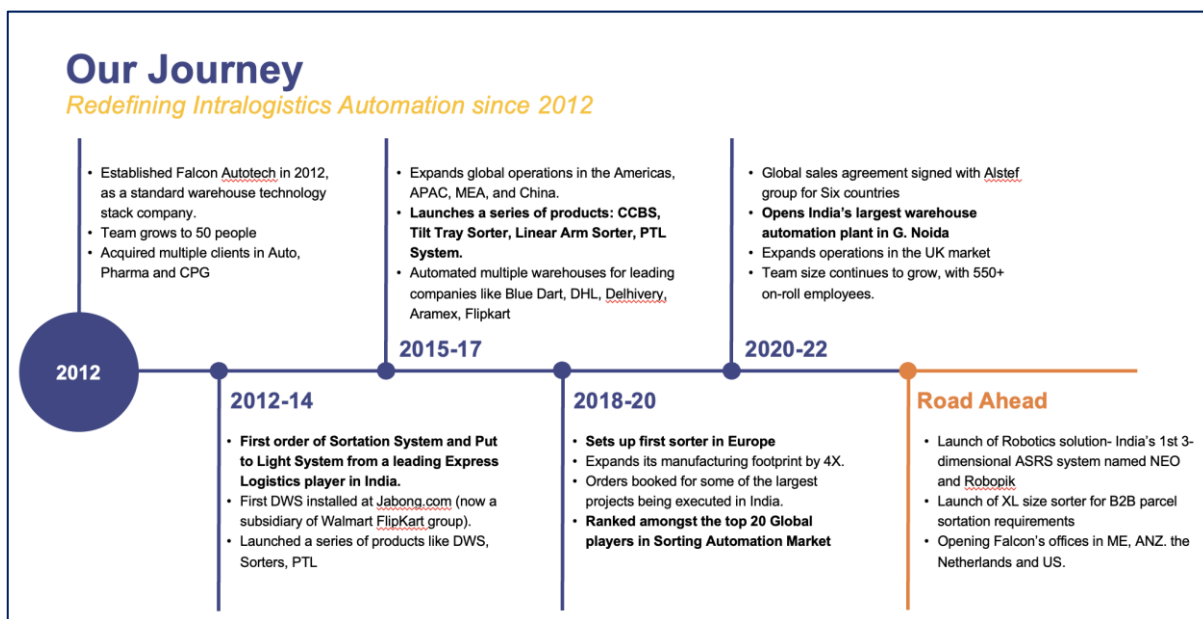
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation companies; our vision is to become the top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. To be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and delivering world-class projects in return.



Product and Solutions

With 100% focused on Parcels, Eatches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- PopUp Sorter
- Sweep Sorter
- Pusher Sorter



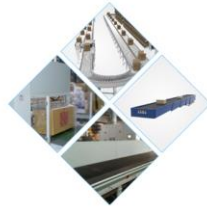
PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
- R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
- Mini (600MM)
- Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

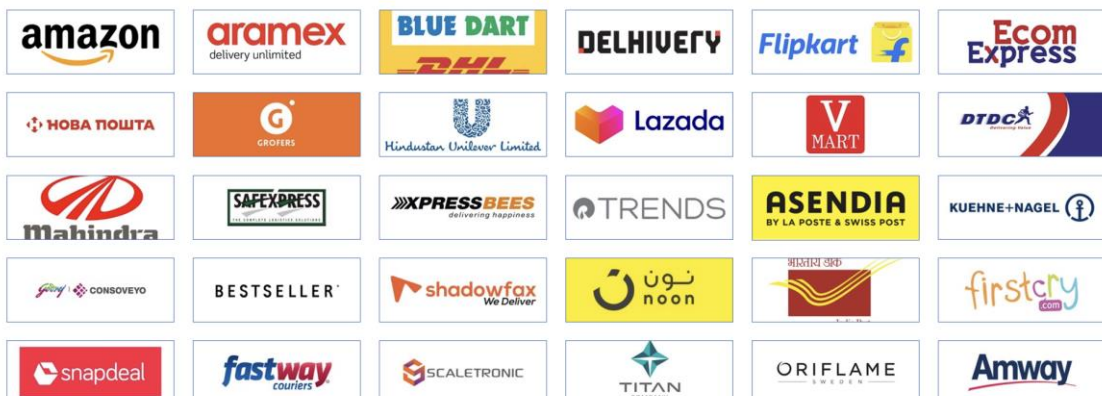


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation, and movement. The process is controlled in real-time by our in-house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions, and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long-standing partnerships. These brands are testimony to our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

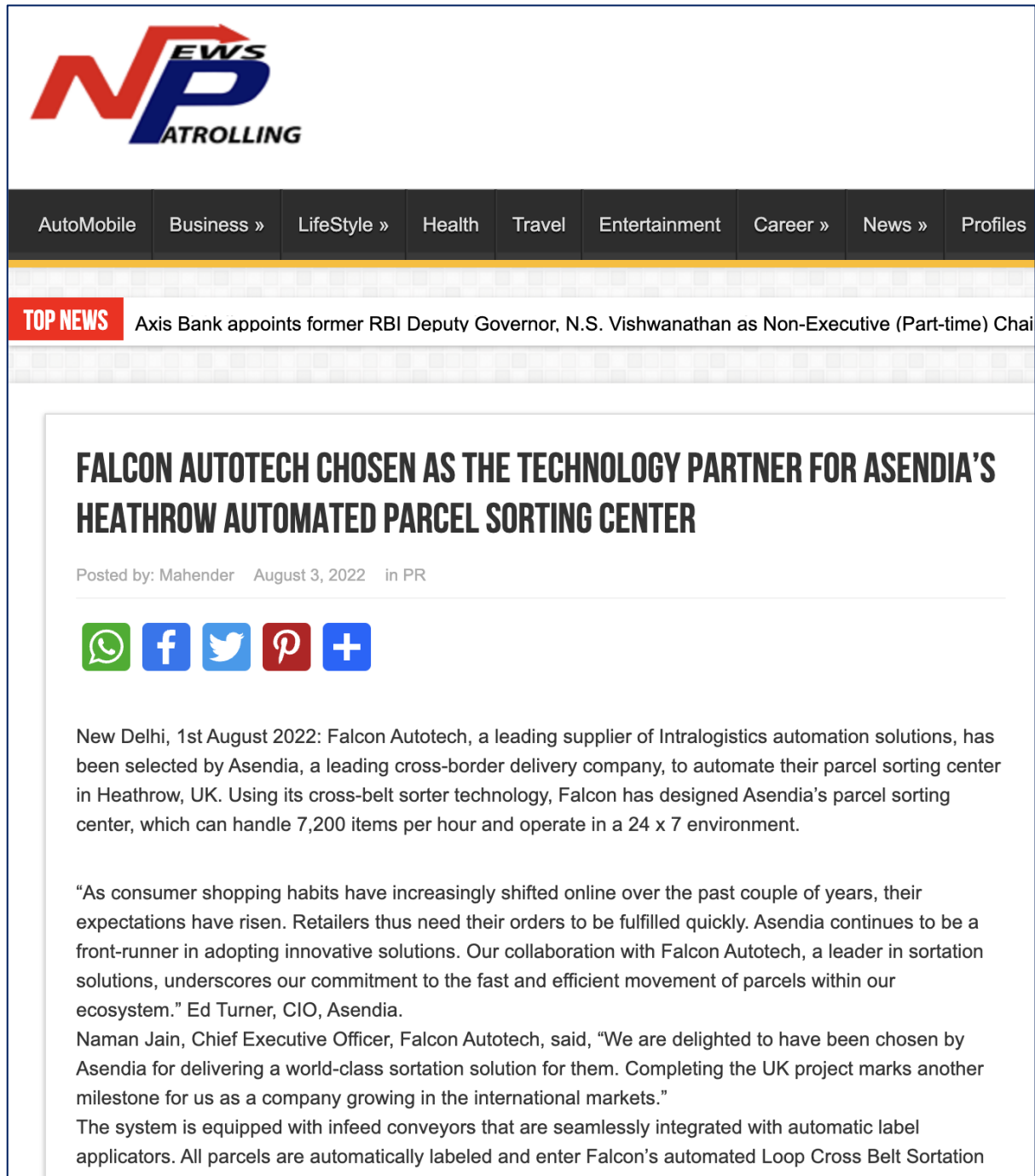


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has a highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chair

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

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New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system.

The parcels are sorted using Falcon's Linear Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –



5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Linear Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

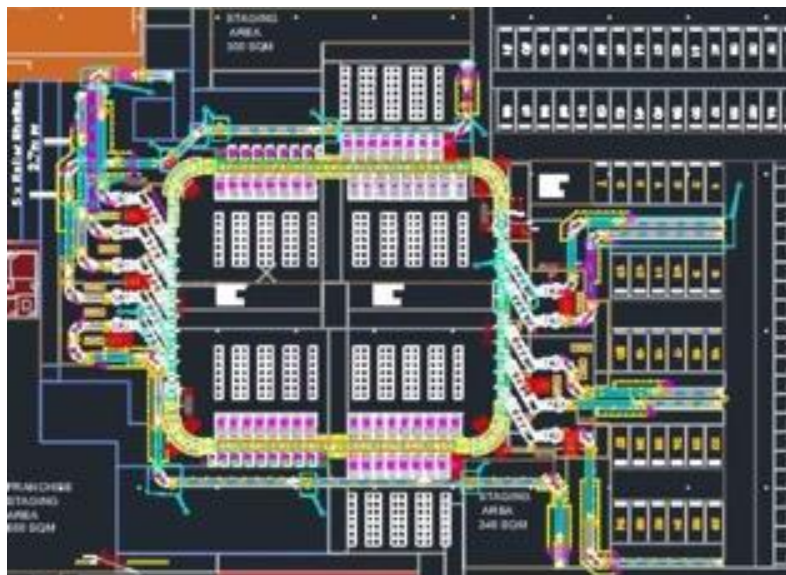
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

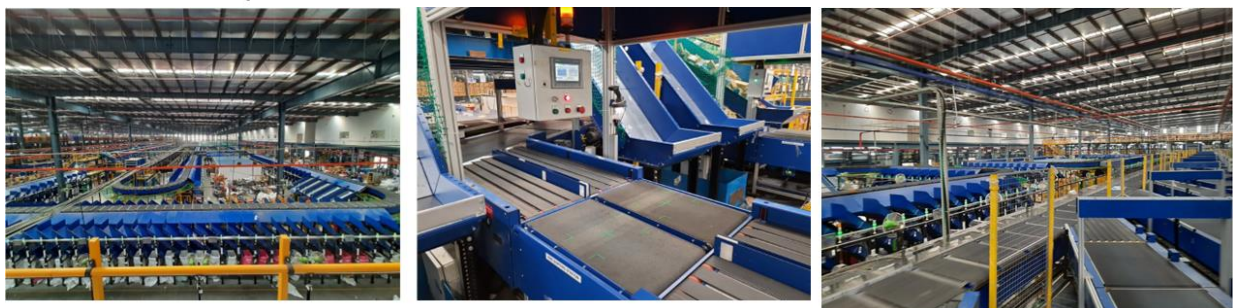
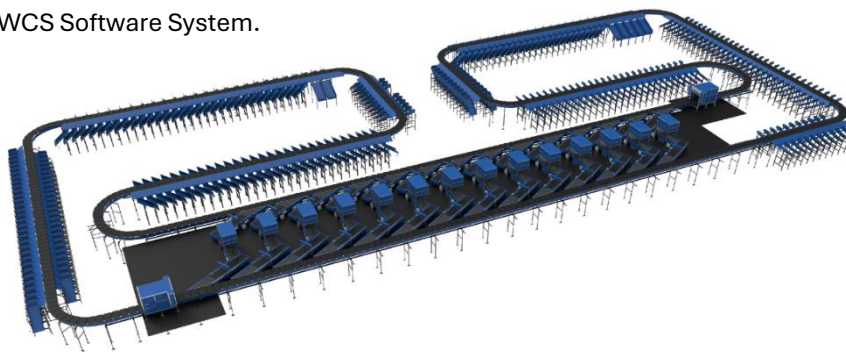
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system.
- WCS Software System.



5.6 Project 6- (Ecommerce Client, India)

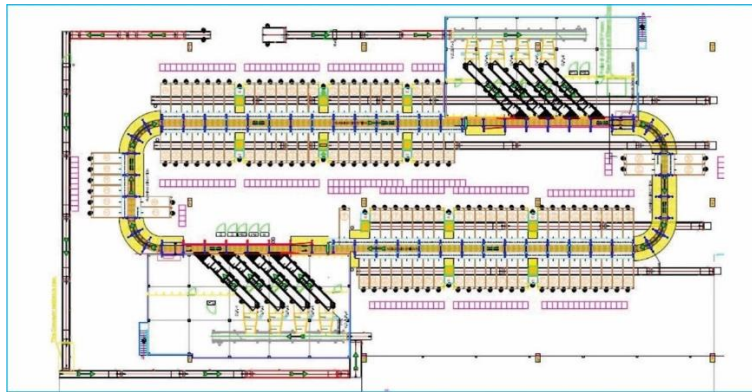
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

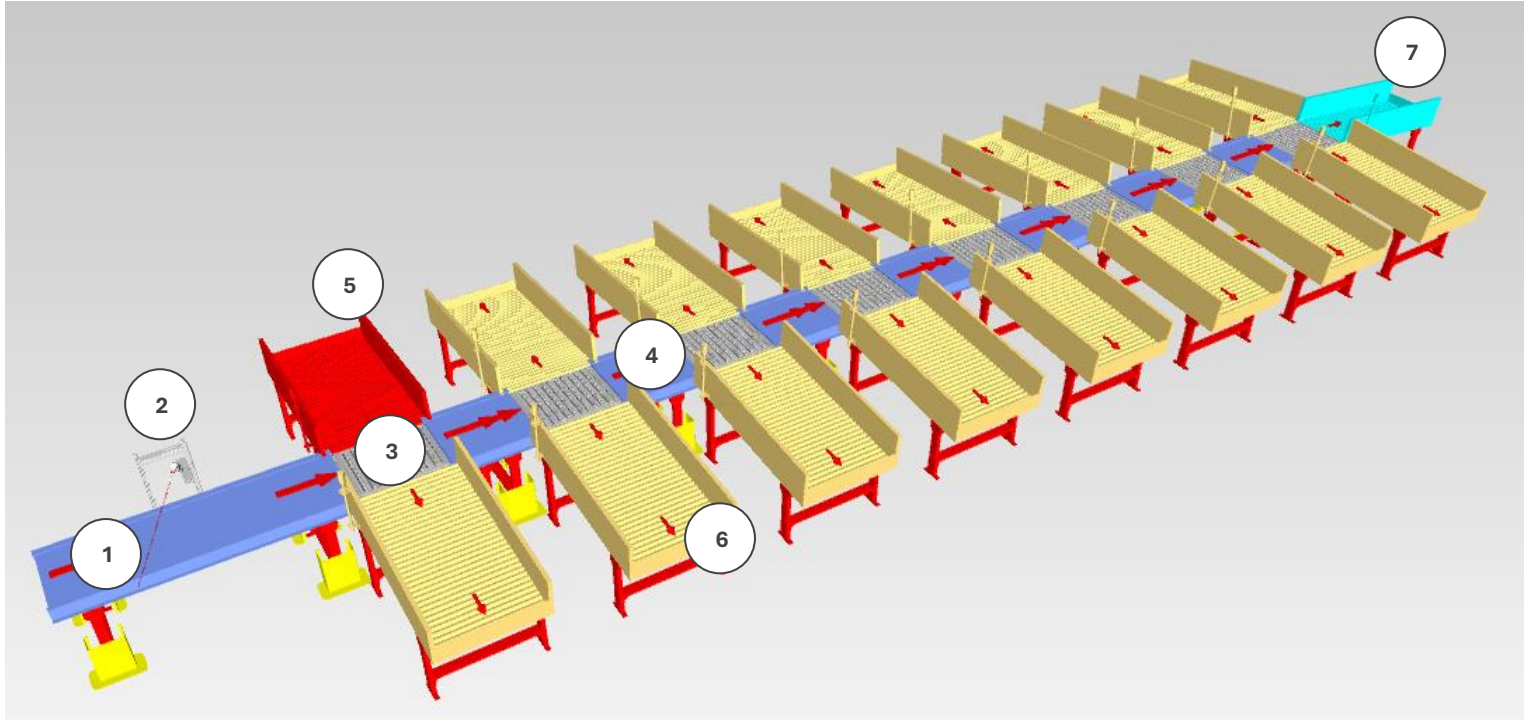
Key Technology Modules - Linear Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



6. Solution Summary & Concept Layout

6.1 Layout View



LEGEND:

1. Loading Conveyor (Powered Roller Conveyor)
2. Side Barcode Scanner
3. Pop-Up Divert Unit
4. Powered Roller Conveyor
5. Rejection Gravity Roller Conveyor
6. Collection Gravity Roller Conveyor
7. Sort-Fail Collection Gravity Roller Conveyor

6.2 Conveyable Product Specifications

Max SKU Length	mm	800
Max SKU Width	mm	600
Max SKU Height	mm	650
Min Length	mm	300
Min Width	mm	200
Min Height	mm	200
Max Weight	Kg	30
Min Weight	grams	100
Type of Product	-	Boxes / Cartons

6.3 Products to be Loaded on Conveyor/Sorter Shall have the following characteristics:

- Centre of Gravity of items must not move during conveyance or sorting.
- Item must not have magnetic content; otherwise, behaviour of parcel cannot be guaranteed.
- Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
- Bags shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
- Plastic ropes shall be perfectly adherent to the surface of the package.
- All items with the risk of being damaged during the transport on an automatic conveying/sorting system or damaging the system. They must be robust enough to avoid disintegration of container material and loose of contents in the conveying/sorting process.
- Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
- Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
- The Bags must have at least one flat and regular surface providing enough stability during conveyance.
- All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.4 Products not to be Loaded on the system:

All products that are not within the range as described here are considered non- conveyable products and must be taken out of the main sorter flow by the operators.

- Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
- Items that have a spherical or cylindrical shape.
- Items that are packed in material that can damage the conveyors or the sorter.
- Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
- Fragile Bags with contents not sufficiently secured.
- Items that have been classified as dangerous are designated.
- Wet items are designated.
- Items with anti-slip treatment.
- Items with protruding parts.
- Items with sharp edges.
- Inadequately packed items that could be damaged during automatic transportation.
- Electrostatically loaded items.
- Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.5 Process Flow

- The Loading operators will receive the packed boxes / cartons in trolleys.
- After Receiving, the operators will load the boxes / cartons on the powered roller conveyor. The operators must ensure that the pre-pasted barcode should on the side of the side of the barcode scanner.
- The side barcode scanner will scan the barcodes on the boxes / cartons.
- Basis the information gathered from the barcode scanning the boxes / cartons will get sorted into their respective collection bins with the help of the pop-up divert units.
- The sort-fail and rejected boxes / cartons will get collected into their respective idler roller conveyor outputs for further processing.

7. System Throughput

Description	Value
Designed Throughput of the system	Up to 800 boxes per hour

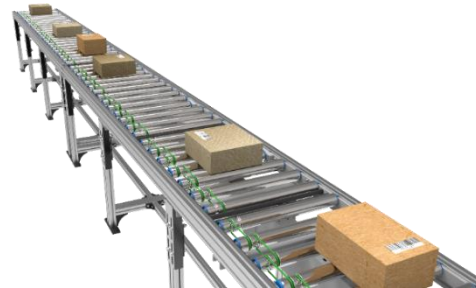
8. System Components

8.1 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	700
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity/unit length of Conveyor	kg/m	50
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	75/100
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

8.2 Side Barcode Scanner

The system is integrated with automated inline side barcode scanning system capable for scanning 1D barcodes with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Cognex/Sick
Barcode Type	Type	1D
Tote Sizes	Mm	Min: 300 x 200 x 200 Max: 800 x 600 x 650
Number of Codes	Nos	1
Location	Side	Left/Right
Orientation	Type	Omnidirectional
Color of bars	Color	Black

8.3 Pop-Up Sorting Unit

Belt Popup device is used in system for 90° transfer of products.

Some Silent Features of Pop-Up:

- Low Noise
- Maximum uptime.
- Electric based popup.
- Minimal maintenance
- High safety standards.



Specification	UOM	Remark
Manufacturer Name	Name	Pulse/Similar Superior
Type – Equipment	Type	Stand Alone
Type – Operation	Type	Yes
Mechanism for Pop-Up	Type	Electric
Drive Power Rating (Lifting)	Watt	40
Type of Motor	Type	DC
Drive Power Rating (For Rollers)	Watt	40
Type of Motor	Type	DC
Ingress Protection type	Type	IP 54
Provision for Carton Box / Carton Detection Sensors	Yes/No	Yes
Load Capacity of Carton Box	Kg	30
Sorting Capacity	PPH	Upto 800

8.4 Gravity Roller Conveyor

Falcon's Roller conveyors require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

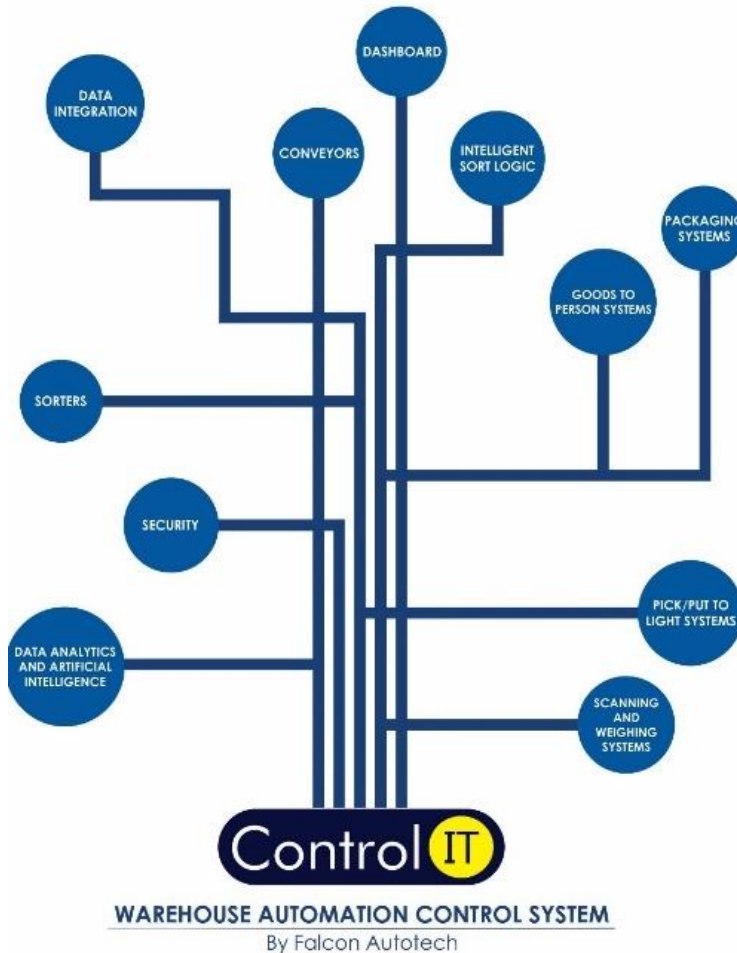
Some Salient Features of Falcon's idler roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	1000
Conveyor Height	mm	As per Layout requirement
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	mm	approx. 600
Roller Diameter	mm	approx. 50
Pitch of Rollers	mm	75
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Yes
End Stopper	yes/no	Yes

9. Falcon's WCS and Control IT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

10. IT infrastructure requirements

Server PC to control the overall system need to be provided by client side as per specifications shared by Falcon Autotech and connectivity of supplied server pc with Client network will be in Client scope.

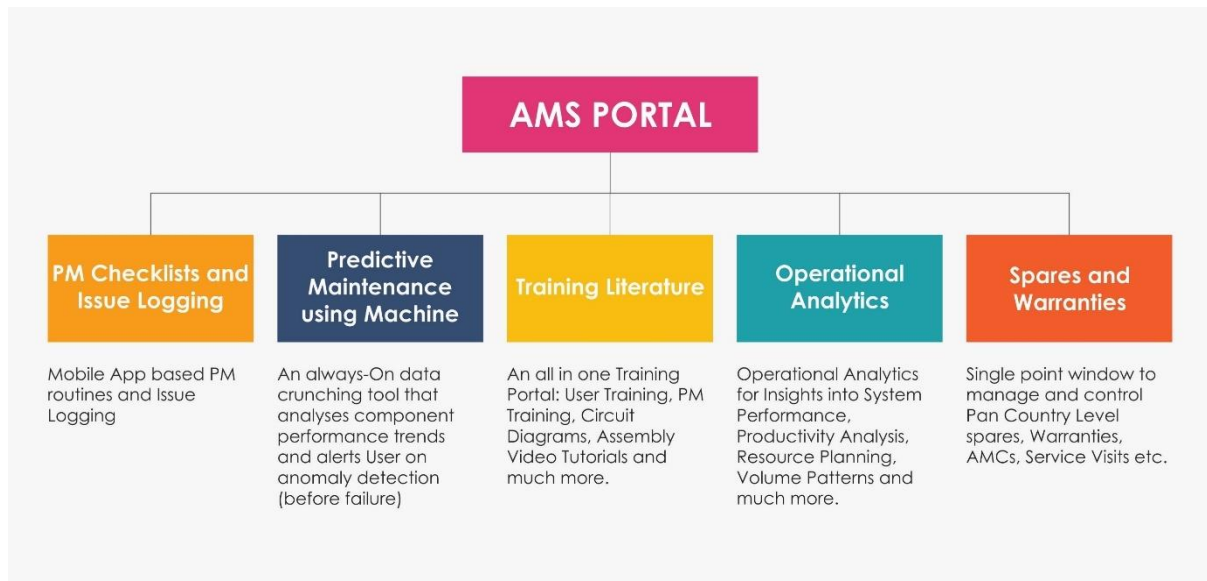
11. Projected productivity benefits with industry benchmarks

1. Integration possibility with conveyors system for post processing like print and apply, weight-based verification, route wise sorting, etc makes it a complete warehouse order processing solution.
2. Sorting Process is completely paperless thereby reducing resources.
3. 99.9% accuracy is achieved in sorting.
4. Maintenance free system with easy troubleshooting.
5. Fast Project Implementation time.

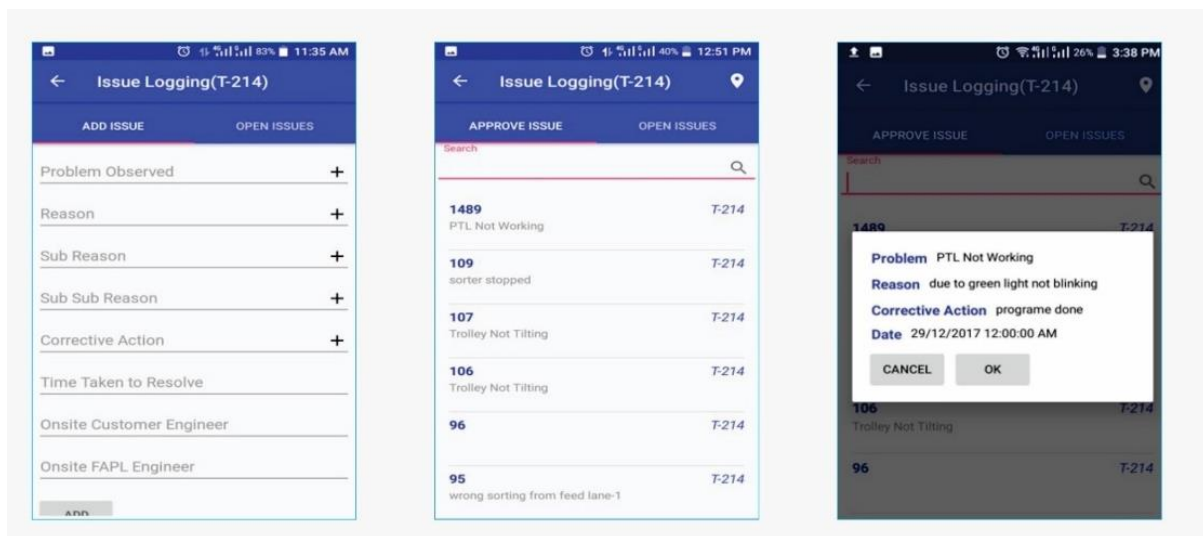
12. Key Components Make

Component	Make
PLC & RIO	Omron
VFD	Omron/Lenze/AB
Ethernet switch	Siemens/Phoenix Contact
SMPS	Omron/Selec/Meanwell
Line chokes	EMIS/GURU NANAK ELE.
Slim relays	Phoenix contact
Terminals	Wago
Phase failure relay	Selec
Cable lugs	Dowell's/TRINITY TOUCH
Utility switch & socket	Anchor
Switchgear	Schneider
Main load break switch (CP)	Schneider
Geared Motors	SEW/Nord
Cameras	Cognex/Sick/Hikvision/IRayple
Hand Scanners	Motorola/Symbol
Belt Conveyor	Falcon Autotech
Pop-Up Divert Unit	Pulse / Itoh Denki/Similar

13. Automation Management System (AMS)



13.1 Issue Logging



13.2 Preventive Maintenance Check List



CL(T-214)
Selected Date : 2018-02-27

HOURLY(4) DAILY(111) WEEKLY MONTHLY

Operations OK
Collection Bin

Chutes overload

Induction NOT OK
Wide Conveyor

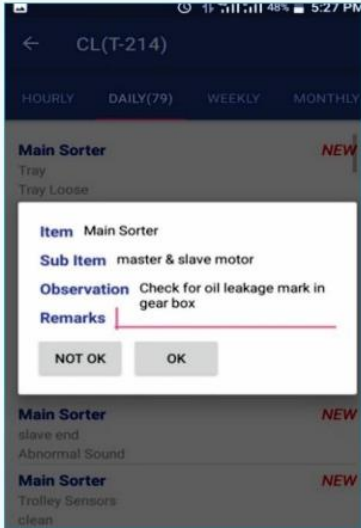
Loading table overload

Rejection NOT OK
Last Collection Bin

Rejection bin overload

PLC Logs OK
PLC

Tray not tilting at last station.



CL(T-214)

HOURLY DAILY(79) WEEKLY MONTHLY

Main Sorter NEW
Tray
Tray Loose

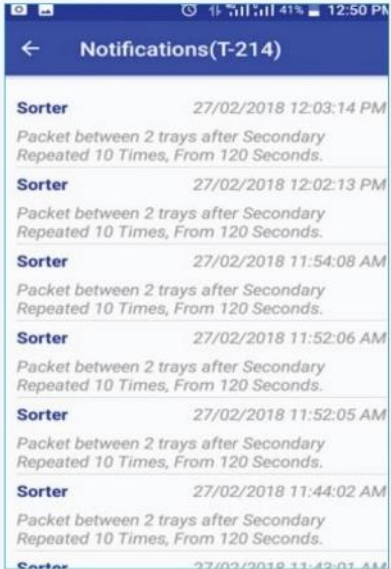
Item Main Sorter
Sub Item master & slave motor
Observation Check for oil leakage mark in gear box
Remarks

NOT OK OK

Main Sorter NEW
slave end
Abnormal Sound

Main Sorter NEW
Trolley Sensors
clean

13.3 Predictive Maintenance



Notifications(T-214)

Sorter 27/02/2018 12:03:14 PM
Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds.

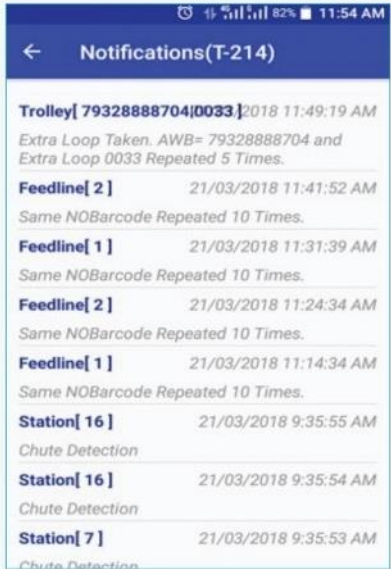
Sorter 27/02/2018 12:02:13 PM
Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:54:08 AM
Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:52:06 AM
Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:52:05 AM
Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:44:02 AM
Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds.



Notifications(T-214)

Trolley[79328888704|0033] 2018 11:49:19 AM
Extra Loop Taken. AWB= 79328888704 and Extra Loop 0033 Repeated 5 Times.

Feedline[2] 21/03/2018 11:41:52 AM
Same NOBarcode Repeated 10 Times.

Feedline[1] 21/03/2018 11:31:39 AM
Same NOBarcode Repeated 10 Times.

Feedline[2] 21/03/2018 11:24:34 AM
Same NOBarcode Repeated 10 Times.

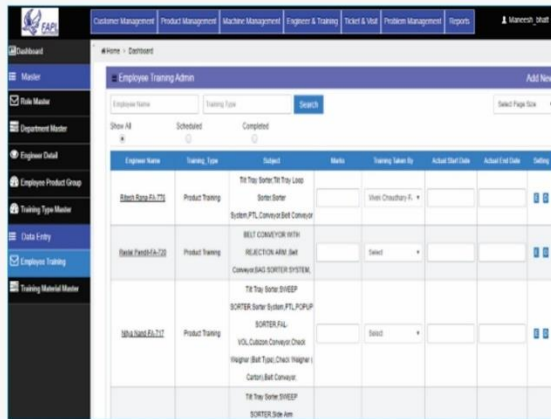
Feedline[1] 21/03/2018 11:14:34 AM
Same NOBarcode Repeated 10 Times.

Station[16] 21/03/2018 9:35:55 AM
Chute Detection

Station[16] 21/03/2018 9:35:54 AM
Chute Detection

Station[7] 21/03/2018 9:35:53 AM
Chute Detection

13.4 Training Portal

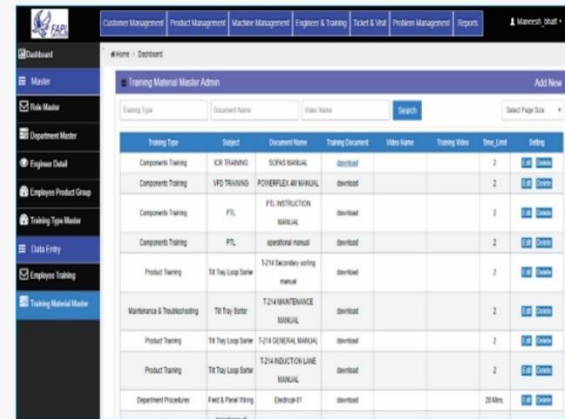


Employee Training Admin

Employee Name: Training Type: Search

Show All Scheduled Completed

Employee Name	Training Type	Subject	Notes	Training Taken By	Actual Start Date	Actual End Date	Rating
Shree Rupa D.771	Product Training	TB Tray Sorter TB Tray Loop Sorter Sorter		Shree Chaitanya K.			
Shree Rupa D.771	Product Training	System PTL Conveyor Belt Conveyor BELT CONVEYOR INTN		Shree Chaitanya K.			
Shree Rupa D.771	Product Training	REJECTION AND SORT CONVEYOR SACS SORTER SYSTEM		Shree Chaitanya K.			
Shree Rupa D.771	Product Training	TB Tray Sorter SHEEP SORTER Sorter System PTL POPUP SORTER FULL		Shree Chaitanya K.			
Shree Rupa D.771	Product Training	YOL Custom Conveyor Check Program (Start Time Check Program) Custom Belt Conveyor		Shree Chaitanya K.			
Shree Rupa D.771	Product Training	TB Tray Sorter SHEEP SORTER Side Arm		Shree Chaitanya K.			



Training Material Admin

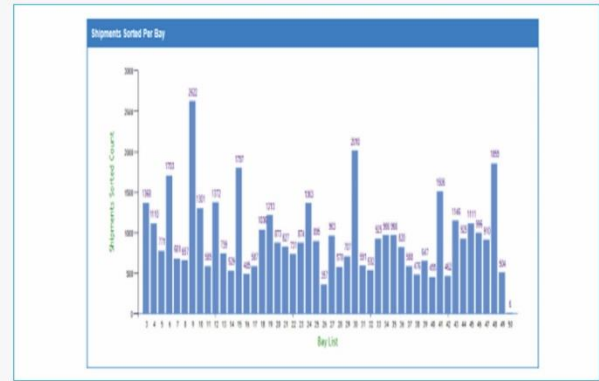
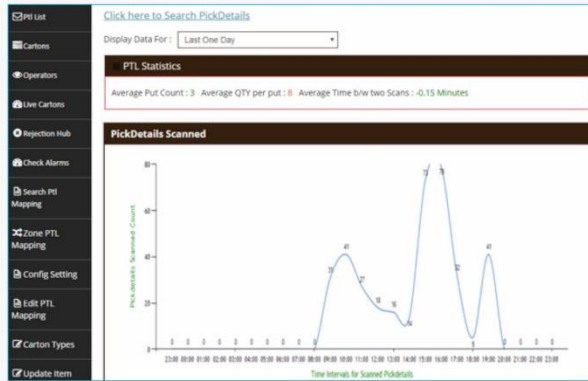
Training Type: Document Name: Video Name: Search

Training Type	Subject	Document Name	Training Document	Video Name	Training Video	Time Limit	Rating
Components Training	CCR SHOOTING	SCPS MANUAL	download			2	100%
Components Training	WPS TRAINING	POWERFLY AIR MANUAL	download			2	100%
Components Training	PTL	PTL INSTRUCTION MANUAL	download			2	100%
Components Training	PTL	operational manual	download			2	100%
Product Training	TB Tray Loop Sorter	T214 Secondary sorting manual	download			2	100%
Maintenance & Troubleshooting	TB Tray Sorter	T214 MAINTENANCE MANUAL	download			2	100%
Product Training	TB Tray Loop Sorter	T214 GENERAL MANUAL	download			2	100%
Product Training	TB Tray Loop Sorter	T214 INDUCTION LANE MANUAL	download			2	100%
Department Procedures	Field & Power Wiring	Electrical PT	download			20 mins	100%

All the mentioned Trainings will be provided on site and training manuals will be shared prior to trainings to the relevant persons.

- Training on how to use the systems, including basic troubleshooting for End users.
- Training on performing day to day system administration activities, maintenance, user, and workflow management for System Administrator(s)
- Training on Solution platform, covering aspects of master configuration and IT users.

13.5 Operational Analysis



14. Proposed System Technical Details / BOM

14.1 Mechanical Equipment

Pos	Qty	Description	Value
FM1	1	Pop-Up Divert Sorter	
		Powered Roller Conveyor	8.9 meters
		Side Barcode Scanner	1 Nos
		Pop-Up Divert Unit	8 Nos
		Collection Gravity Roller Conveyor	15 Nos
		Rejection Gravity Roller Conveyor	1 Nos
		Sort-Fail Gravity Roller Conveyor	1 Nos
		Hand-Barcode Scanner for Semi-Automatic Mode	1 Nos

14.2 Electrical Equipment

Electricals			
FC1	1	Electrical Cabling and Wiring	
		Main Power Distribution Panel	
		Main Control Panel	
		Sorter Drive Controls	
		Network Switches	
		Field Cabling	
			Included

14.3 Control System

Components			
FC2	1	Controls and IT Infrastructure Controls	
		Barcode Scanning Controls	
		Conveyor Controls	
		PLC Based Controls	
		IT	
		Control IT and Software Packages	
		Networking Switches	
			Included

15. System Warranty, Service Level, Onsite Support Package and Spare

Package (Optional)

15.1 System Standard Warranty

Falcons offered System comes with a standard warranty of 1 year and post completion of warranty, customer can opt from Various Maintenance Packages.

Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

Note - Shipping charges of Faulty Spare Back to Falcon, Packing, Forwarding, Insurance, Freight, Duties, Taxes etc are payable by customer.

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

15.2 Requesting Service

- Email: support@falconautoonline.com
- Telephonic: Support Manager Contact Number will be shared post Hand Over
- Requesting Hours: 24 hours

- Office Working Days: Mondays to Saturdays (Excluding Indian national holidays. List can be shared upon request)

15.3 Service Response

Considering the Installation site being at Farukhnagar, Gurugram, India,

First Response:

- Within 2 Hours of Issue Logging between Monday to Saturday (Excluding Indian National Holidays and Public Holidays) and within 4 hours of issue logging for rest of the days.

Engineer Physical Site Visit:

- Within 6-12 Hours of Identification for a need of engineer visit during office hours (09 A.M. to 05:00 P.M.)
- Within 12-24 Hours of Identification for a need of engineer visit post office hours (05 P.M. to 09:00 A.M.).

Mean Time to Repair (MTTR):

- With Spares and Engineer/Trained Sorter Operator available on site, mean time to make the system up will be 2 hours calculated on Monthly Basis.

15.4 Uptime Commitment

Falcon commits 99% uptime of its supplied system with the following considerations:

1. Complete Loss of throughput shall be deemed as downtime.
2. Any downtime caused due to factors not attributable to Equipment or Staff of Falcon would be excluded.
3. On Site Support (OSS) will be available for all shifts.
4. Spares shall be available on site.
5. 2 Hours of Continuous Preventive Maintenance downtime shall be provided daily.
6. Runtime 21 hours/day

15.5 Falcon's Spares Package

Falcon suggests its clients to buy and keep a stock of recommended spares or at least critical spares on site such that in case of any part failure, the part from the client's stock can be utilized to make the system up and the utilized part can be replenished by Falcon Autotech Free of Cost in case the system is under warranty.

16. Comprehensive Annual Maintenance Package (Optional)

16.1 Engineering Support Package

The Engineering Support Package covers the following:

- Telephonic Support
- Email Support
- Remote Assistance via Team Viewer or Any other remote session software.
- Software Upgrades & Bug Fixes (New IT Developments are excluded).
- Physical On-Site Assistance

16.2 Spares Package

- General Spare replacement/repair in case of faulty spare.
- Cost of Spares in Falcon Autotech Scope.
- Replacement of Spares will be subject to failure analysis and approval for Falcon Autotech.
- Shipping charges of Faulty Spare Back to Falcon, Packing, Forwarding, Insurance, Freight, Duties, Taxes etc are payable by customer.

16.3 Consumables & Belts Package

- The supply, replacement, and installation of any worn-out/ wear & tear items such as belts, O-rings, bearings, etc during the tenure of the contract
- The shipping charges of the consumables shall be borne by Falcon.

Note –

- Printer Consumables such as Printer Ribbon and Paper Roll are Excluded and need to be purchased by customer as per consumption needs.

16.4 General Terms & Conditions

- System entering contract should be in a good working condition. A health checkup of the system could be done by the OEM/Channel Partner team.
- Training, Data Backup, Relocation of Machine, Cabling, Wiring, OS Support, Application Support and Virus Support will not be included under service contract.
- Any damage to the equipment due to factors such as Operation, Mechanical Damage, Damage due to electrical fluctuations/Surges/ etc., damage due to fire or any other damage that cannot be attributed to OEM/Falcon Autotech Shall not be covered under service contract. In Such cases

Falcon Autotech will take all reasonable steps to ascertain the RCA in coordination with the OEM and final decision of OEM shall be binding on buyer.

- The scope of extended warranty would exclude any damage caused due to physical, mechanical, electrical or fire failure by the user. The Repair/Replacement of the equipment and the engineer visits in such a case would be done on chargeable basis.

16.5 Customer Responsibilities

- Proper reporting of the Issue faced via the contact routes specified.
 - a. For this it is recommended that customer should train and authorize their employee(s) in coordination with FAPL (SELLER)
 - b. Trained customer employee(s) shall be available on site during each intervention (remote and on site); this(these) employee(s) shall also apply safety and rescue measures in case of emergency.
- Data transmission requirements for tele and remote support (data stream, internet connection speed etc.) – the customer shall ensure the sufficient quality of the telephone and remote connection; any delays and additional costs caused by insufficient data transmission or lack of cooperation from the on-site personnel shall be borne by the customer.
- The availability of all parts which require customer-specific programming and configuration (e.g., barcode scanner, 3D camera, wireless router, PLCs) within the agreed reaction times for on-site interventions is not guaranteed; the delivery times for these parts shall be requested separately; the customer shall ensure the on-site availability of these parts to avoid delays for the on-site interventions.
- Perform all relevant checks and maintenance activities according to the user and maintenance manual which are not in the scope of the Service contract.
- Training to Machine Operators and technicians
- Maintaining, occupational health, safety of on-site manpower and environmental protection
- To send any damaged parts with proper packaging and documentation back to FAPL, Greater Noida-India, facility for RCA, wherever required.
- Parts purchased from FAPL must be used and maintained by customer.
- In no event shall SELLER be liable to BUYER for any consequential, indirect, special, or punitive damages, including but not limited to loss of business, goodwill, income, profit or savings as well as any other claim for consequential compensation arising out of or relating to this AGREEMENT.
- The BUYER is not authorized/allowed to replace, adjust, repair or in any way manipulate the automated system and its components unless he is specially trained and authorized to perform the corresponding manipulation.
- All damages to SELLER's equipment caused by force, negligence, improper use, improper replacement, adjustment, and manipulation shall be borne by the BUYER.

17. Commercial

S/N	Components	Unit	Qty	Price (INR)
1	Pop-Up Divert Units	Nos	8	₹ 36,41,849
2	Side Barcode Scanner	Nos	1	
3	Powered Roller Conveyor	meters	8.9	₹ 18,22,197
4	Gravity Roller Conveyor (2 meters)	Nos	17	
5	Field Wiring & Controls	set	1	₹ 6,57,857
6	Software & IT Integration	set	1	₹ 6,00,000
7	Project Management & Installation	set	1	₹ 4,75,643
Grand Total				₹ 71,97,545
Discount				₹ 11,97,545
Final Price				₹ 60,00,000

18. Payment terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: Release of PO – 40%

Stage 2: 50% along with 100% Taxes against Dispatch on Pro-rata Basis –

Stage 3: Post Installation – 10%

Note – All Payments to be done within 30 Days of Invoice or getting due.

19. Terms & Conditions

Total Price is to be understood:

1. Ex-Works Greater Noida
2. Excluding Taxes & Duties
3. Including Installation Charges
4. Excluding the Material Unloading and Laydown area at customer's site.
5. Excluding Freight.

20. Program Organisation

20.1 Project Schedule

16-18 weeks ex-works from the date of PO and Advance.

During the project execution, any delay which cannot be attributed to Falcon shall be excluded from the calculation and a corresponding extension shall be provided to Falcon.

20.2 Documentation

The following documents will be prepared and submitted:

- Requirements Document
- User manual
- Training manual
- Installation and operations manual

20.3 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Client
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.

21. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Server PC
- Onsite Amenities Like Washroom, Water, etc to be provided during installation to the installation team.
- Safety Netting, barricading, Bollards, toe guards or any such safety requirements shall be in Clients Scope.
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.